

Experiment 07

Configuration and Use of Network Devices

Lab Objective:

To **configure** and **understand the functions** of various **network devices** such as **routers, switches, and hubs**, and learn their role in a network setup.

Prerequisites:

- Basic understanding of **networking concepts** (e.g., IP addressing, data transmission).
- Familiarity with **network configuration commands** and tools (e.g., Cisco Packet Tracer, command line interface, IP configuration).

Outcome:

By the end of this experiment, students will be able to:

- Identify different **network devices** (router, switch, hub).
- Understand how they function in a network.
- Successfully **configure** routers and switches using **commands/tools**.
- Demonstrate data flow through a simple network.

1. Hub

- **Definition:** A hub is a **basic networking device** that connects multiple computers in a LAN.
- **Function:** It simply **broadcasts data** to all ports, regardless of the destination.
- **Type:** Physical Layer device (Layer 1 of OSI model).

- **Working:** When a data packet arrives at one port, it is copied to all other ports.
- **Configuration:**
No configuration needed — plug and play.
- **Advantages:**
 - Cheap and easy to use.
- **Disadvantages:**
 - No intelligence (no MAC filtering).
 - Causes network congestion.
- **Use Case:** Small or temporary networks for testing.

2. Switch

- **Definition:** A switch is a networking device that **connects devices** in a LAN and uses **MAC addresses** to forward data only to the intended device.
- **Function:** Makes decisions based on the **destination MAC address**.
- **Layer:** Data Link Layer (Layer 2).
- **Working:**
 - Maintains a **MAC address table**.
 - Learns the devices connected to each port and sends data to the correct port.
- **Basic Configuration (using CLI):**

```
enable
configure terminal
hostname Switch1
interface FastEthernet0/1
switchport mode access
switchport access vlan 10
exit
```

- **Advantages:**
 - Reduces network traffic.
 - More secure than hubs.
- **Disadvantages:**
 - More expensive than hubs.
- **Use Case:** Office or campus LAN environments.

3. Router

- **Definition:** A router is a device that **connects multiple networks** and forwards data packets between them based on IP addresses.
- **Function:** Routes data between **different networks** (e.g., LAN to WAN).
- **Layer:** Network Layer (Layer 3).
- **Working:**
 - Uses **routing tables** and **routing protocols** (RIP, OSPF) to determine the best path.
- **Basic Configuration (using CLI):**

```
enable
configure terminal
hostname Router1
interface FastEthernet0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
interface FastEthernet0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
exit
```

- **Advantages:**
 - Enables communication between different networks.
 - Provides firewall and security features.
- **Disadvantages:**
 - More complex and expensive.
- **Use Case:** Internet access, connecting LANs to WANs.

 Comparison Table:

Device	OSI Layer	Intelligent ?	Forwards by	Configuration Needed	Use Case
Hub	Layer 1	No	Broadcasts	No	Small testing networks
Switch	Layer 2	Yes	MAC address	Basic VLAN setup	LAN communication
Router	Layer 3	Yes	IP address	IP Routing setup	Internet / WAN connectivity

Sample Network Setup:

Let's assume a simple network:

- **2 PCs** connected to a **switch**.
- **Switch** connected to a **router**.
- Router connects to another network or Internet.

Steps:

1. Assign IP addresses to PCs (e.g., 192.168.1.2 and 192.168.1.3).
2. Configure switch (optional VLAN).
3. Configure router interfaces to connect internal and external networks.
4. Test connectivity using ping command.

Conclusion:

Understanding and configuring network devices like **hubs, switches, and routers** is essential for building and maintaining a functional network. These devices ensure proper **data delivery, efficiency, and security** in communication.